

Exponentials & Logarithms



Laws of logarithms

- 1 Simplify $\log_2 8 + \log_2 4$ to a single number.
 - 2 Express $2\ln x - \ln(x - 1)$ as a single logarithm.
 - 3 Simplify $\log_3 54 - \log_3 2$ to a single number.
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Solving exponential equations using logarithms

- 4 Solve $5^x = 20$, giving your answer to three decimal places.
 - 5 Solve $3^{2x-1} = 10$, to three decimal places.
 - 6 Solve $7^{3x} = 50$, to three decimal places.
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Solving logarithmic equations

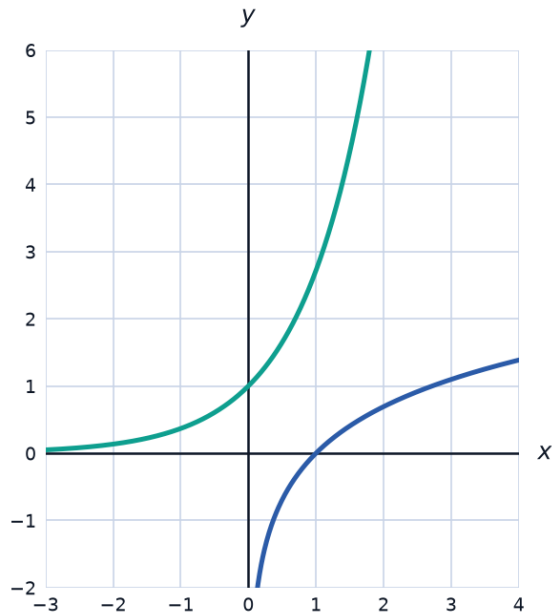
- 7 Solve $\ln(2x - 1) = 3$, to three decimal places.
 - 8 Solve $\log_2(x + 3) + \log_2(x - 1) = 5$, rejecting any value outside the domain.
 - 9 Solve $\ln x + \ln(x + 3) = \ln 10$, rejecting any value outside the domain.
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Properties of the exponential and natural log functions

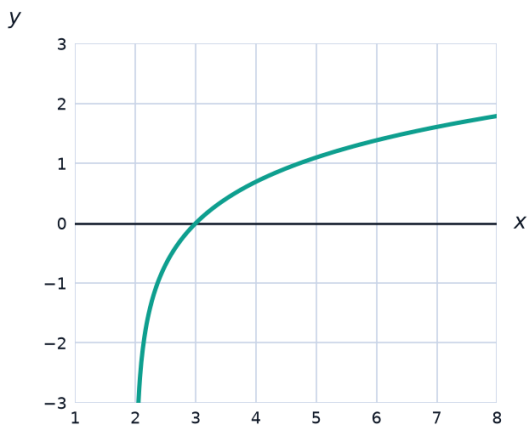
- 10 Simplify $e^{\ln 7}$ and $\ln(e^4)$, using the fact that e^x and $\ln x$ are inverse functions.
 - 11 Solve $e^{2x} = 15$, to three decimal places.
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Graphs of exponential and logarithmic functions

- 12 The graph shows $y = e^x$ and $y = \ln x$. State how the two curves are related geometrically, and state the equation of the asymptote of each.



- 13 The graph shows $y = \ln(x - 2)$. State its domain and the equation of its vertical asymptote.



Exponential growth and decay models

- 14 A bacteria culture grows according to $N(t) = 200e^{0.15t}$, where t is in hours. Find the population after 10 hours, to the nearest whole number.
- 15 Using the same model $N(t) = 200e^{0.15t}$, find how long it takes the population to double, to three decimal places.

Simultaneous equations involving exponentials and logarithms

- 16 Solve simultaneously $\log_2 x + \log_2 y = 5$ and $x - y = 4$, for $x, y > 0$.
- 17 Solve simultaneously $\ln x - \ln y = 1$ and $x + y = 11$, giving x and y to three decimal places.
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The change of base formula

- 18 Evaluate $\log_5 30$ using the change of base formula, to three decimal places.
- 19 Solve $2^x = 3^{x-1}$ using logarithms, to three decimal places.
- 20 Evaluate $\log_7 3 \times \log_3 7$ without a calculator, explaining why the product is exactly 1.
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A well-designed word problem: radioactive decay

- 21 This question has three parts. A radioactive isotope decays according to $A(t) = A_0 e^{-kt}$, where A_0 is the initial mass (grams) and t is time in years. (a) Given that the isotope has a half-life of 12 years, find the decay constant k , to four decimal places. (b) A sample starts with $A_0 = 80$ grams. Find the mass remaining after 30 years, to two decimal places. (c) Find how long it takes the sample to decay to 5 grams.